Lu Sun, and many more.

A Notebook on Graph Theory and Topology

To my family, friends and communities members who have been dedicating to the presentation of this notebook, and to all students, researchers and faculty members who might find this notebook helpful.

Contents

Foreword

I was not a big supporter for sharing notebooks. Notebooks, in my view, had been private and practically useless to anyone other than their owner.

It traces back to the summer of 2009 when I started my third year as a high school student. In the end of August when the results of Gaokao (National College Entrance Examination of China, annually held in July) came out, photocopy shops would start selling notebooks photocopies that they claimed to be from the top scorers of the exam. Much curious as I was about what these notebooks look like, never have I purchased any of them, and nor expected myself to actually learn anything from them. In fact, many of these notebooks were more difficult to read than the textbooks. I guess we could not blame the top scorers for being so smart that they sometimes made things extremely brief or overwhelmingly comprehensive. Not to mention, why would I want to adapt to notebooks of others when I had my own notebooks which in my opinion should be just as good as theirs.

However, my mind changed after becoming an undergraduate student in 2010 with the school of electrical engineering. There were so many engineering modules, and as an unfortunate result it was very challenging to dig deep into all of them evenly. I naturally put less effort on "side" modules such as material science than my major's core modules such as analog electronics. The situation became worse when I started pursuing my Ph.D. in 2014. As I had to focus on specific research areas entirely, I could hardly split enough time to systematically study those less-relevant yet interesting contents. For example, my research area was on robust statistics, state estimation and adaptive control, and I could not spend as much time as I wished on graph theory. I needed a way to quickly scratch the surface of some areas, and became decently familiar with them. There was no intention to become an expert on these areas by all means. I just wanted to understand what they are, and see whether they could have inspired my own research.

To make a difference, I forced myself to read articles beyond my comfort zone and it ended up motivating me to take notes to consolidate the knowledge. I started with easy subjects. My very first notebook was on Numerical Analysis, an entrance-level module for engineering background graduate students. Revisiting that module did not bring much benefit to my research, but it served as a good starting point. Following that were some more math notebooks, then a few on computing, operating systems and databases, and then on data science, and then on ontology.

Eventually at one point, it suddenly came to me: these notebooks that I had taken while learning these subjects as an armature, they were exactly the materials I needed, quoting myself, to "quickly scratch the surface of some areas, and became decently familiar with them". Well, in that case, why not digitizing them, making them accessible online and open-source and letting everyone read and edit it?

Hence, this is where we are.

Similar with most open-source software, these notebooks do not come with any "warranty" of any kind, meaning that there is no guarantee that everything in the notebooks is correct. Always bear in mind that the notebooks are not peer reviewed. **Do NOT cite this notebook or any notebooks in this series in your academic research paper!** If you find anything helpful here and you want to use it in your research, please trace back to the origin of the knowledge and confirm by yourself.

This notebook is suitable as:

- a quick reference guide;
- a brief introduction for beginners of an area;
- a "cheat sheet" for students or teachers for an entrance-level course.

This notebook is NOT suitable as:

- a direct research reference;
- a replacement of the corresponding textbook.

Again, this notebook is NOT peer reviewed. It is meant to be easy to read, not to be comprehensive and very rigorous.

Although this notebook is open-source, the reference materials of this notebook, including textbooks, journal papers, conference proceedings, etc., may not be open-source. Very likely many of these reference materials are licensed or copyrighted. Please legitimately access these materials and properly use them, should you decided to trace the origin of the knowledge.

Some of the figures in this notebook are plotted using Excalidraw, a very convenient tool to emulate hand drawings. The Excalidraw project can be found on GitHub, *excalidraw/excalidraw*. Other figures may come from MATLAB, R, Python, and other computation engines. The source code to reproduce the results are likely to be included in the same repository of the notebook, but there might be exceptions.

This work might have benefited from the assistance of large language models, which are used exclusively for editing purposes such as correcting grammar and rephrasing sentences, without introducing new content, generating innovative information, or changing the original intent of the text.

Preface

This notebook discusses graph theory and general topology.

A list of key references of this notebook is given below. These materials are frequently cited in the notebook. For convenience, they are not given in the Reference section but listed here instead, whereas otherwise I will have to cite them everywhere in the text.

Topology

- Willard, Stephen. General topology. Courier Corporation, 2012.
- MIT Open Course, Introduction To Metric Spaces

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General Topology

1

Set Theory and Metric Space

CONTENTS

Set theory and metric space are introduced in this chapter as the preliminaries to general topology. They are also the fundamentals to many other subjects such as abstract algebra, topological analysis and real analysis.

1.1 Metric Space

Metric space is introduced in this section.

1.1.1 Definition

A **metric space** is a non-empty set X with a distance function $d: X \times X \to \mathbb{R}_{\geq 0}$, where function d satisfies the following properties.

- Symmetric: $\forall x, y \in X,\ d(x,y) = d(y,x)$;
- Positive definite: $\forall x, y \in X,\ d(x,y) \geq 0$, and $d(x,y) = 0$ if and only if $x = y$;
- Triangle inequality: $\forall x, y, z \in X,\ d(x,z) \geq d(x,y) + d(y,z)$.

Is metric space an algebraic system?

No, we do not consider metric space an algebraic system.

The distance function defined in a metric space is essentially a mapping but not an operator. The tools and properties we study on metric spaces and algebraic systems are also different.

In that perspective, though both metric space and algebraic system define set and "rules" of some kind, they are different.

Consider the following metric space (X, d), where the distance function is defined by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & \text{otherwise} \end{cases} \tag{1.1}$$

It can be easily proved that the above meets the criteria of metric space, hence (X, d) a metric space. This is probably one of the simplest metric spaces and it demonstrates that metric space can be defined on any non-empty set.

1.1.2 Commonly Seen Metric Spaces

Euclidean distance (also known as **l^2-distance**) is a metric space defined on $\mathbb{R}^n$, with the distance function defined by

$$\|x - y\|_2 = \left(\sum_n |x_i - y_i|^2\right)^{\frac{1}{2}} \tag{1.2}$$

Euclidean distance, which be frequently used in the remainder of this notebook, is one of the most popular metric spaces in subjects such as real analysis and topological analysis.

Another example of metric space is the **l^∞-distance** on $\mathbb{R}^n$ which is defined by

$$\|x - y\|_\infty = \max_n |x_i - y_i| \tag{1.3}$$

It can be easily verified that both Euclidean distance and l^∞-distance on $\mathbb{R}^n$ satisfy the definition of metric space.

Both Euclidean distance and l^∞-distance can be derived from **p-norm** which is defined on $\mathbb{R}^n$ as follows.

$$\|x\|_p = \left(\sum_n x_i^p\right)^{\frac{1}{p}} \tag{1.4}$$

Letting $p = 2$ in (1.4) gives

$$\|x\|_2 = \left(\sum_n x_i^2\right)^{\frac{1}{2}}$$

which is known as the **Euclidean norm**, and Euclidean distance in (1.2) is the Euclidean norm of $x - y$.

Letting $p \to \infty$ in (1.4) gives

$$\|x\|_{\infty} = \max_{n} |x_i|$$

which is known as the **infinity norm**, and l^{∞}-distance in (1.2) is the infinity norm of $x - y$.

In this sense, p-norm can be taken as a kind of metric space.

Metric space has many use cases and are used in many definitions of more abstract concepts. Some examples are given as follows.

In the context of metric space, define **ball of radius r centered at p** as follows.

$$B_r(p) = \{x \in X | d(x,p) < r\}$$

With that defined, open set can be defined as follows. A set $A \subseteq X$ is an **open set** if $\forall a \in A$, $\exists \epsilon > 0$ such that $B_{\epsilon} a \subseteq A$.

There are different ways to define a continuous function. The ϵ-δ definition which uses metric space is given below.

Let (X, d_x), (Y, d_y) be two metric spaces defined on sets X and Y respectively, and let $f : X \to Y$ be a function mapping elements from X to Y. Function f is a **continuous function** if $\forall \epsilon > 0$, $\exists \delta > 0$ such that

$$d(x_a, x_b) < \delta \implies d\left(f(x_a), f(x_b)\right) < \epsilon$$

In other words, the difference of the function values can be made arbitrarily small if the input independent variables are close.

Let $f, g \in C^0\left([a,b]\right)$ be two continuous functions on interval $[a,b]$, $f, g : [a,b] \to \mathbb{R}$. Let X be a set containing such continuous functions, and define the distance function

$$d(f,g) = \sup_{x \in [a,b]} |f(x) - g(x)|$$

where $f, g \in X$. It can be proved that (X, d) is a metric space. This is demonstrated by Fig. 1.1, where the red and blue curve are two continuous functions and their distance on $[a,b]$ is given by the black solid segment.

The proof of (X, d) to be a metric space can be done fairly easily using the definition, and it is neglected here.

The same idea applies to multivariable functions. It can also extend from continuous function C^0 to continuously differentiable functions C^1 (the function is differentiable and its differential is also continuous), in which case the distance function becomes

$$d(f,g) = \sup_{x \in [a,b]} |f(x) - g(x)| + \sup_{x \in [a,b]} |f'(x) - g'(x)|$$

and to n-th order continuously differentiable function C^n, in which case the distance function becomes

$$d(f,g) = \sum_{i=0}^{n} \sup_{x \in [a,b]} |f^{(i)}(x) - g^{(i)}(x)| \tag{1.5}$$

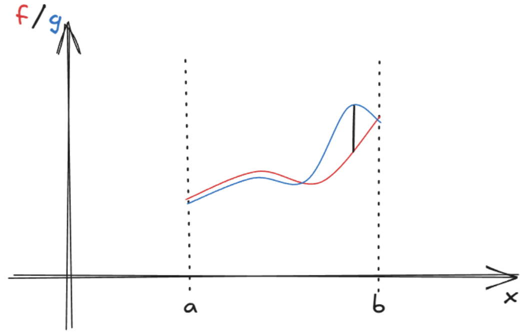

FIGURE 1.1: A demonstrative example of a metric space defined on continuous functions.

As for infinity-order continuously differentiable function C^∞ which is also known as the **smooth function**, (1.5) might be unbounded. Practically, we shall either limit the use of the metric to only bounded-distance smooth functions, or make a walk around with modified distance functions such as

$$d(f,g) = \sum_{i=0}^{\infty} 2^{-i} \frac{\sup_{x\in[a,b]}|f^{(i)}(x) - g^{(i)}(x)|}{1+\sup_{x\in[a,b]}|f^{(i)}(x) - g^{(i)}(x)|}$$

Bibliography